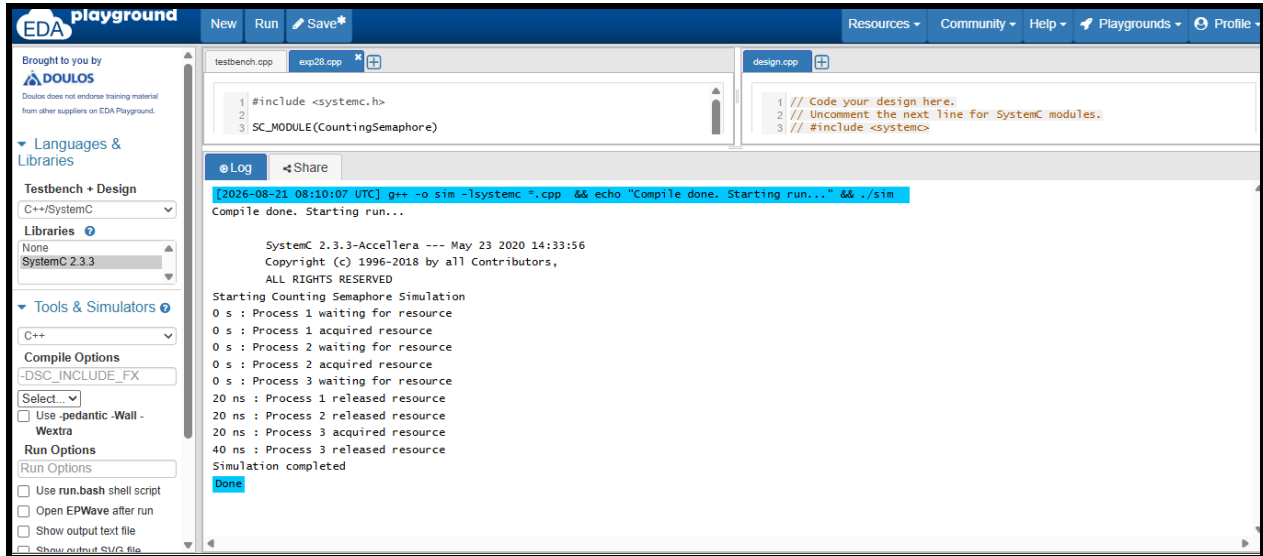




## C++ Code:

```
#include <systemc.h>

SC_MODULE(CountingSemaphore)
{
    sc_semaphore sem;

    void process1()
    {
        cout << sc_time_stamp() << " : Process 1 waiting for resource" << endl;

        sem.wait();

        cout << sc_time_stamp() << " : Process 1 acquired resource" << endl;

        wait(20, SC_NS);

        cout << sc_time_stamp() << " : Process 1 released resource" << endl;

        sem.post();
    }

    void process2()
    {
        cout << sc_time_stamp() << " : Process 2 waiting for resource" << endl;

        sem.wait();

        cout << sc_time_stamp() << " : Process 2 acquired resource" << endl;
    }
}
```

```

        wait(20, SC_NS);

        cout << sc_time_stamp() << " : Process 2 released resource" << endl;

        sem.post();
    }

    void process3()
    {
        cout << sc_time_stamp() << " : Process 3 waiting for resource" << endl;

        sem.wait();

        cout << sc_time_stamp() << " : Process 3 acquired resource" << endl;

        wait(20, SC_NS);

        cout << sc_time_stamp() << " : Process 3 released resource" << endl;

        sem.post();
    }

    SC_CTOR(CountingSemaphore)
        : sem("sem", 2)
    {
        SC_THREAD(process1);
        SC_THREAD(process2);
        SC_THREAD(process3);
    }
};

int sc_main(int argc, char* argv[])
{
    CountingSemaphore cs("CountingSemaphore");

    cout << "Starting Counting Semaphore Simulation" << endl;

    sc_start(100, SC_NS);

    cout << "Simulation completed" << endl;

    return 0;
}

```